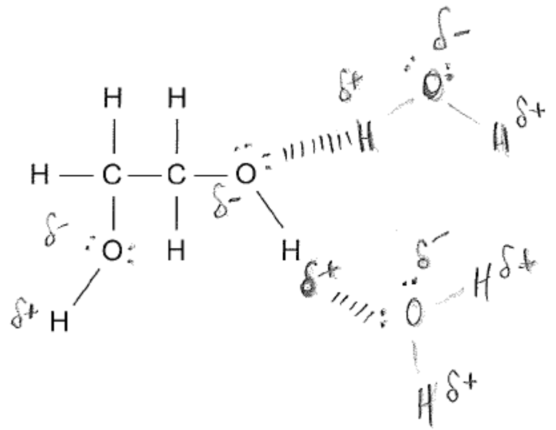
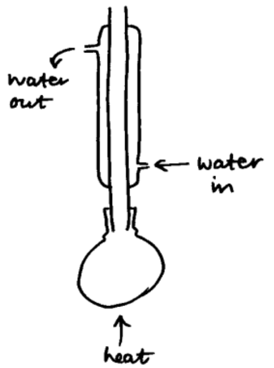
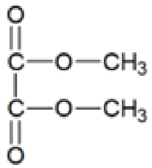
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13.	(a)	(i)		 dipole shown on O—H bond using δ+ and δ– (either water **or** diol) (1) lone pair of electrons shown on oxygen atom (either water **or** diol) (1) hydrogen bond shown between diol oxygen (lone pair) and hydrogen atom in water **or** between water oxygen (lone pair) and appropriate hydrogen atom in diol (1)		3		3		
		(ii)		award (1) for any of following • hydrogen bonding interferes with the formation of the ice lattice • hydrogen bonding stops the ice lattice from forming • disrupts the hydrogen bonding between water molecules • water molecules hydrogen bond more strongly with diol molecules than each other • water prefers to form hydrogen bonds with the diol than with itself neutral answer – hydrogen bonds form between molecules of the diol and water molecules			1	1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		award (1) for either of following acidified (potassium) dichromate or $Cr_2O_7^{2-}$ / H^+ acidified (potassium) manganate(VII) or MnO_4^- / H^+	1			1		1
		(ii)		accept formulae in any format e.g. $CH_2OHCH_2OH + 4[O] \rightarrow (COOH)_2 + 2H_2O$ $(CH_2OH)_2 + 4[O] \rightarrow (COOH)_2 + 2H_2O$ $C_2H_6O_2 + 4[O] \rightarrow C_2O_4H_2 + 2H_2O$ correct formulae of reagents and products (1) balancing (1)		2		2		
		(iii)		 vertical condenser with flask being heated (1) water going through condenser in correct direction (1) award (1) only if condenser shown in distillation apparatus with water going from low to high	2			2		2

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(iv)		$(HC{=}O)_2^+$ / $(HCO)_2^+$ / $H_2C_2O_2^+$ must have + charge do not accept 2+ charge			1	1		
		(v)		moles of ethane-1,2-diol = $\frac{2.00}{62.06}$ = 0.0322 (1) this gives 0.0323 mol of ethanedioic acid M_r of ethanedioic acid = $\frac{3.94}{0.0322}$ = 122 (1) relative mass of water = 122 – 90 = 32 (1) ⇨ x = 1.78 / 1.8 / 2 (1) **Alternative method** moles of ethane-1,2-diol = $\frac{2.00}{62.06}$ = 0.0322 (1) this gives 0.0323 mol of ethanedioic acid 0.0322 × 90 = 2.90 g (1) 3.94 – 2.90 = 1.04 g of water moles of water = $\frac{1.04}{18}$ = 0.0577 (1) mole ratio 0.0322 : 0.0577 1:1.79 ⇨ x = 1.79 / 1.8 / 2 (1)		4		4	3	

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(vi)	I	 + $2H_2O$ full structure of the diester (1) two molecules of water (1)	1	1		2		
			II	ester	1			1		
				Question 13 total	**5**	**10**	**2**	**17**	**3**	**3**